MCS Saturday SAT Math

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Digital SAT Math · Systems of Linear Equations

Systems Reasoning · Homework

New Digital Format · Algebra · Activity Sheet

Name _____ **Date** _____

Completion Check Show a valid setup, solve efficiently, and verify each correction.

A Read the System

6 prompts · 10 min

Do not solve unless asked. Read structure first: slope, intercept, coefficients, and equation relationships.

A1 · Classification Reason

The equations **3x + 2y = 14** and **6x + 4y = 28** form what kind of system? State the number of solutions and the structural evidence.

Infinitely many solutions; a consistent, dependent system.
The second equation is exactly 2 times the first: 2(3x + 2y = 14) gives 6x + 4y = 28.

A2 · Parameter Analyze

For what value of **k** will **kx + 6y = 15** and **4x + 8y = 20** represent the same line? Explain using a common scale factor.

k = 3.
Multiply 4x + 8y = 20 by 3/4 to get 3x + 6y = 15.
All three terms use the same scale factor.

A3 · Coordinate Meaning Explain

A system has solution **(7, −2)**. Write the two substitution checks that must both be true for this ordered pair to solve the system.

For equations ax + by = c and dx + ey = f, both checks must hold:
7a − 2b = c and 7d − 2e = f.
Substitute x = 7 and y = −2 in each equation.

A4 · Fast Recognition Audit

Without graphing, explain why **y = −3x + 5** and **6x + 2y = 4** cannot intersect.

6x + 2y = 4 becomes y = −3x + 2.
Both slopes are −3, but the y-intercepts are 5 and 2. The lines are parallel and distinct, so they never intersect.

A5 · Target Quantity Translate

A question asks for **twice the x-coordinate** of the intersection. If the system solution is **(a,b)**, what is the required response? Explain why **b** is irrelevant.

2a.
The x-coordinate is a; b is the y-coordinate and is not part of the requested quantity.

A6 · Representation Connect

Write a second equation that, with **2x − 5y = 9**, creates a system with no solution. Your equation may not be identical to the first.

Example: 2x − 5y = 10.
The left sides are identical but the constants differ, so no ordered pair can satisfy both.

B Choose the Cleanest Door

4 decisions · 7 min

Select the most efficient first move, then give one sentence of justification.

B1 Level 2

$y = 5x - 9$ and **$3x + 2y = 19$**

A. Substitution
B. Elimination
C. Graphing
D. Guess and check

Why: _____

A. Substitution.
Since y is already isolated, replace y in the second equation with $5x - 9$:
$3x + 2(5x - 9) = 19$.

B2 Level 2

$7x - 4y = 1$ and **$7x + 3y = 29$**

A. Substitution
B. Elimination
C. Graphing
D. Table

Why: _____

B. Elimination.
Subtract the first equation from the second to cancel x immediately:
$7y = 28$.

B3 Level 2

Two equations contain decimals and the prompt asks for the approximate intersection shown in a graphing context.

A. Substitution
B. Elimination
C. Desmos intersection
D. Mental math

Why: _____

C. Desmos intersection.
Graph both equations and inspect their intersection coordinates to obtain the requested approximate solution without cumbersome decimal arithmetic.

B4 Level 2

The coefficients of **x** are 5 and 15, while no variable is isolated.

A. Isolate y first
B. Multiply one equation, then eliminate x
C. Graph by hand
D. Average coefficients

Why: _____

B. Multiply one equation, then eliminate x.
Multiply the equation whose x-coefficient is 5 by 3, then subtract to cancel the matching 15x terms.

C Build, Solve, Verify

6 problems · 16 min

For each problem, show the key algebraic step. Check the ordered pair in both original equations.

C1 Level 1

What is the solution to the system **x + y = 11** and **x − y = 3**?

A. (4, 7) **B.** (7, 4)
C. (8, 3) **D.** (11, 3)

B. (7, 4).
Add the equations: 2x = 14, so x = 7.
Then y = 11 − 7 = 4.
Checks: 7 + 4 = 11; 7 − 4 = 3.

C2 Level 1

The system **y = 2x + 1** and **y = −x + 10** has solution **(x,y)**. What is **y**?

SPR: _____

y = 7.
2x + 1 = −x + 10 ⇒ 3x = 9 ⇒ x = 3.
Then y = 2(3) + 1 = 7.
Checks: 7 = 2(3) + 1; 7 = −3 + 10.

C3 Level 2

What is the **x-coordinate** of the solution to **4x + 3y = 1** and **2x − 3y = 17**?

SPR: _____

x = 3.
Add: 6x = 18 ⇒ x = 3.
6 − 3y = 17 ⇒ y = −11/3.
Checks: 12 − 11 = 1; 6 + 11 = 17.

C4 Level 2

For the system **5x + 2y = 16** and **3x − 2y = 8**, what is the value of **x + y**?

A. 2 **B.** 3
C. 4 **D.** 5

x + y = 7/2 = 3.5. No listed choice is correct.
Add: 8x = 24 ⇒ x = 3.
15 + 2y = 16 ⇒ y = 1/2.
Checks: 15 + 1 = 16; 9 − 1 = 8.

C5 Level 3

The graphs of **2x − y = 7** and **ax + y = 5** intersect at **(3,−1)**. What is the value of **a**?

SPR: _____

a = 2.
3a − 1 = 5 ⇒ 3a = 6 ⇒ a = 2.
Checks at (3, −1):
2(3) − (−1) = 7; 2(3) + (−1) = 5.

C6 Level 3

The solution to **3x + by = 17** and **x − y = 1** is **(5,4)**. What is **b**?

A. −1/2 **B.** 1/2
C. 2 **D.** 8

B. b = 1/2.
3(5) + 4b = 17 ⇒ 4b = 2 ⇒ b = 1/2.
Checks: 15 + (1/2)(4) = 17; 5 − 4 = 1.

D Translation Lab

4 models · 12 min

Define both variables, write the system, solve, and answer the exact question with units.

D1 · Tickets Model

Adult tickets cost $18 and student tickets cost $11. A group buys 26 tickets for $349. How many student tickets were purchased?

Let a = adult tickets and s = student tickets.
a + s = 26
18a + 11s = 349
Substitute a = 26 − s:
18(26 − s) + 11s = 349
468 − 7s = 349 ⇒ s = 17; a = 9.
Answer: 17 student tickets.
Check: 9 + 17 = 26; 162 + 187 = $349.

D2 · Mixture Model

A lab needs 30 liters of a 42% solution by mixing a 30% solution with a 50% solution. How many liters of the 50% solution are needed?

Let x = liters of 30% solution and y = liters of 50% solution.
x + y = 30
0.30x + 0.50y = 0.42(30) = 12.6
Substitute x = 30 − y:
9 + 0.20y = 12.6 ⇒ y = 18; x = 12.
Answer: 18 liters of 50% solution.
Check: 12 + 18 = 30 L; 3.6 + 9 = 12.6 L of solute.

D3 · Accounts Model

Two savings accounts contain $8,400 total. One earns 4% annually and the other 6%. Together they earn $438 in one year. How much is in the 6% account?

Let x = dollars at 4% and y = dollars at 6%.
x + y = 8,400
0.04x + 0.06y = 438
Substitute x = 8,400 − y:
336 + 0.02y = 438 ⇒ y = 5,100.
Then x = 3,300.
Answer: $5,100 in the 6% account.
Check: 3,300 + 5,100 = 8,400;
132 + 306 = $438 interest.

D4 · Data Plan Model

Plan A costs $24 plus $8 per gigabyte. Plan B costs $54 plus $3 per gigabyte. At how many gigabytes do the plans cost the same, and what is that common cost?

Let g = gigabytes and c = common cost in dollars.
c = 24 + 8g
c = 54 + 3g
Set equal: 24 + 8g = 54 + 3g.
5g = 30 ⇒ g = 6.
c = 24 + 8(6) = 72.
Answer: 6 gigabytes; common cost $72.
Check: 24 + 48 = 72; 54 + 18 = 72.

E Two-Pass Systems Set

10 questions · 18 min

Pass 1: collect direct points. Pass 2: return to parameter and modeling items. Leave no response blank.

Name the target | Exploit structure | Use Desmos deliberately | Verify in context

E1 Level 1

If **y = x + 6** and **y = 4x − 3**, what is the value of **x**?

A. 1 **B.** 2
C. 3 **D.** 4

C. x = 3.
x + 6 = 4x − 3 ⇒ 9 = 3x ⇒ x = 3.
Check: y = 9 in both equations.

E2 Level 1

What is the **y-coordinate** of the solution to **x + 2y = 13** and **x = 5**?

SPR: _____

y = 4.
5 + 2y = 13 ⇒ 2y = 8 ⇒ y = 4.
Check: 5 + 2(4) = 13; x = 5.

E3 Level 2

Which ordered pair solves **2x + 5y = 4** and **x − y = 9**?

A. (7, −2) **B.** (8, −1)
C. (9, −2) **D.** (−7, 2)

A. (7, −2).
x = y + 9 ⇒ 2(y + 9) + 5y = 4.
7y = −14 ⇒ y = −2, x = 7.
Checks: 14 − 10 = 4; 7 − (−2) = 9.

E4 Level 2

The system **6x + 9y = 12** and **2x + 3y = k** has infinitely many solutions. What is **k**?

SPR: _____

k = 4.
Divide 6x + 9y = 12 by 3:
2x + 3y = 4.
The second equation must be the same line.

E5 Level 2

A store sold 32 notebooks. Plain notebooks cost $4 and premium notebooks cost $7. The total revenue was $170. How many premium notebooks were sold?

A. 10 **B.** 12
C. 14 **D.** 18

C. 14 premium notebooks.
p + q = 32; 4p + 7q = 170.
128 + 3q = 170 ⇒ q = 14; p = 18.
Check: 18 + 14 = 32; 72 + 98 = $170.

E6 Level 3

Lines **ax + 4y = 12** and **3x − 2y = 5** are parallel. What is the value of **a**?

SPR: _____

a = −6.
Slopes: −a/4 and 3/2.
−a/4 = 3/2 ⇒ a = −6.
Intercepts 3 and −5/2 differ, so the lines are distinct and parallel.

The timed set contains new questions, not the attached lesson questions.

E Two-Pass Systems Set · continued

10 questions · 18 min

Pass 1: collect direct points. Pass 2: return to parameter and modeling items. Leave no response blank.

Name the target | Exploit structure | Use Desmos deliberately | Verify in context

E7 Level 3

The solution to **px + y = 13** and **2x − y = 2** has x-coordinate 5. What is **p**?

A. 1/5 **B.** 6/5
C. 11/5 **D.** 3

p = 1. No listed choice is correct.
With x = 5, 2(5) − y = 2 ⇒ y = 8.
Then 5p + 8 = 13 ⇒ p = 1.
Checks: 1(5) + 8 = 13; 10 − 8 = 2.

E8 Level 3

For a positive constant **m**, the system **mx + 3y = 15** and **4x + 6y = 18** has no solution. What is **m**?

SPR: _____

m = 2.
Divide the second equation by 2:
2x + 3y = 9.
For no solution, the first must have the same left side but a different constant:
2x + 3y = 15 versus 2x + 3y = 9.

E9 Level 4

A company sells small boxes for $12 and large boxes for $20. On one day it sold 75 boxes. Revenue from large boxes was $380 more than revenue from small boxes. How many large boxes were sold?

SPR: _____

40 large boxes.
Let s = small boxes and l = large boxes.
s + l = 75; 20l − 12s = 380.
20l − 12(75 − l) = 380
32l = 1,280 ⇒ l = 40; s = 35.
Check: 35 + 40 = 75;
20(40) − 12(35) = 800 − 420 = $380.

E10 Level 4

In the system **(r+1)x + 2y = 8** and **6x + (r−1)y = 12**, there is no unique solution when the coefficient determinant is 0. What is the **positive** value of **r**?

A. √10 **B.** √11
C. √12 **D.** √13

D. r = √13.
Determinant = (r + 1)(r − 1) − 2(6).
Set it to zero: $r^2 - 1 - 12 = 0$.
$r^2 = 13 \Rightarrow r = \pm\sqrt{13}$.
The positive value is √13.

F Process Evidence

After the timer · 6 min

Record observable evidence—not a general feeling about difficulty.

F1 Reflect

Which item did you flag first, and what exact feature made it expensive?

Sample reflection: I flagged E10 first because both variable coefficients depend on r, and expanding the determinant required more setup than a direct substitution.

F2 Reflect

Where did elimination save steps compared with substitution?

Sample reflection: In C3, adding the equations canceled +3y and −3y immediately, giving 6x = 18 without isolating a variable.

F3 Reflect

If you used Desmos, what feature did you inspect? If not, where would it have helped?

Sample reflection: I did not use Desmos. In D4, it could show the intersection (6, 72), confirming the break-even usage and cost.

F4 Reflect

Which answer did you verify in both equations or in context? Show that check.

Sample reflection: I verified D1 in context:
9 + 17 = 26 tickets.
18(9) + 11(17) = 162 + 187 = $349.
Both the ticket count and cost match.

H First Broken Line

4 forensic checks · 8 min

For any missed item, identify the earliest line at which the solution path stopped being valid.

H1 Audit

Problem number and first invalid line

Sample error, C3: After correctly getting x = 3, I wrote 6 − 3y = 17, then −3y = 23. The first invalid line is −3y = 23.

H2 Audit

Why that line looked reasonable at the time

Sample explanation: I added 6 instead of subtracting 6 from both sides. The arithmetic looked plausible, but it did not preserve equality.

H3 Audit

A replacement line that preserves equivalence

Sample correction:
6 − 3y = 17 ⇒ −3y = 17 − 6 = 11.
Thus y = −11/3; the solution is (3, −11/3).

H4 Audit

A fast check that would expose the error

Sample check: The erroneous y = −23/3 gives 2(3) − 3(−23/3) = 29, not 17.
Correct checks: 12 − 11 = 1 and 6 + 11 = 17.

I Error Log

Up to 6 corrections

Use one primary error type: Knowledge, Process, Reading, Timing, or Blank/Unreached.

#	Problem	My original thought	Primary error type	Correction rule
1	C3 (sample)	I changed $6 - 3y = 17$ to $-3y = 23$.	Process	Subtract 6 from both sides: $-3y = 11$.
2	A1 (sample)	Different-looking equations must intersect once.	Knowledge	Compare all terms: exact multiples mean the same line.
3	A5 (sample)	I reported b instead of twice the x-coordinate.	Reading	Name the target first: the requested value is 2a.
4	G2 (sample)	I treated all $5,520 as revenue before upgrades.	Reading	Subtract the $240 upgrade revenue before solving.
5	E10 (sample)	I expanded $(r + 1)(r - 1)$ as $r^2 + 1$.	Process	Use difference of squares: $r^2 - 1$; then $r^2 - 13 = 0$.
6	G3 (sample)	I accepted both roots of the zero determinant.	Knowledge	For infinitely many solutions, compare constants too; only $k = 5$ works.

J Transfer Reflection

Complete sentences

The system feature I now notice before calculating is...

Sample: The system feature I now notice before calculating is whether coefficients are equal, opposite, or proportional.

The cleanest method-switch I made was... because...

Sample: The cleanest method-switch I made was from substitution to elimination in C3 because the y-terms canceled immediately.

One parameter problem became simpler when I compared...

Sample: One parameter problem became simpler when I compared all coefficient and constant ratios in G3, not just the determinant.

On my next Digital SAT module, I will verify a systems answer by...

Sample: On my next Digital SAT module, I will verify a systems answer by substituting into both original equations and checking the requested quantity and units.

Bring the completed sheet and error log to the next session.

G Bonus Challenge

3 Level 4 + 1 Level 5 · untimed

These are extension problems. Show a defensible path; a correct answer without supporting work is incomplete.

G1 Level 4

For real constants **a** and **b**, the system **ax + 3y = 7** and **4x + by = 10** has solution **(2,1)**. What is the value of **a+b**?

A. 2 **B.** 3
C. 4 **D.** 5

C. a + b = 4.
Substitute (2, 1) into both equations:
2a + 3 = 7 ⇒ a = 2.
8 + b = 10 ⇒ b = 2.
Therefore a + b = 2 + 2 = 4.
Checks: 2(2) + 3(1) = 7;
4(2) + 2(1) = 10.

G2 Level 4

A theater sold 240 tickets. Balcony tickets cost $18 and orchestra tickets cost $30. After 20 balcony-ticket purchases were upgraded to orchestra tickets for an additional $12 each, total ticket revenue was $5,520. How many tickets were **originally** orchestra tickets?

SPR: _____

80 originally orchestra tickets.
Let b = original balcony tickets and o = original orchestra tickets.
b + o = 240
18b + 30o + 20(12) = 5,520
18(240 − o) + 30o = 5,280
12o = 960 ⇒ o = 80; b = 160.
After upgrades: 140 balcony, 100 orchestra.
Check: 140(18) + 100(30) = $5,520.

G3 Level 4

The system **(k−2)x + 3y = 9** and **6x + (k+1)y = 18** has infinitely many solutions. What is **k**?

SPR: _____

k = 5.
For infinitely many solutions, all coefficients and constants must share one scale factor.
Since 18/9 = 2, the second equation must be twice the first.
6 = 2(k − 2) ⇒ k = 5.
Also k + 1 = 2(3) ⇒ k = 5.
Check: 3x + 3y = 9 doubles to 6x + 6y = 18.
The other zero-determinant value, k = −4, does not make the constants proportional.

G4 Level 5

For constants **p** and **q**, every solution of **2x + py = 8** also satisfies **qx + 6y = 24**. The point **(1,t)** lies on both lines. What is the value of **p+q+t**?

SPR: _____

p + q + t = 11.
Both equations describe the same line.
The constant ratio is 24/8 = 3, so multiply the first equation by 3:
6x + 3py = 24.
Compare with qx + 6y = 24:
q = 6 and 3p = 6 ⇒ p = 2.
At (1, t): 2 + 2t = 8 ⇒ t = 3.
Thus p + q + t = 2 + 6 + 3 = 11.
Check: 2 + 2(3) = 8; 6 + 6(3) = 24.